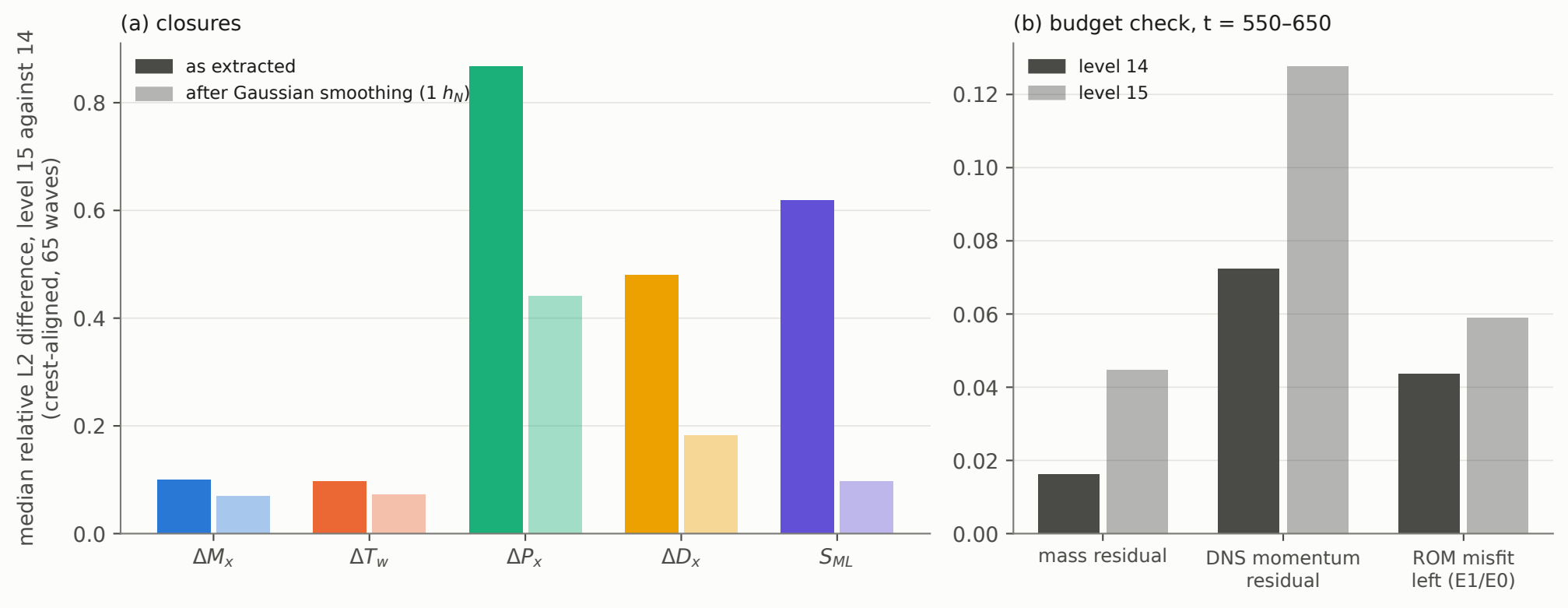

(a) closures
as extracted
after Gaussian smoothing (1 h_N)
median relative L2 difference, level 15 against 14
(crest-aligned, 65 waves)
0.0
0.2
0.4
0.6
0.8
ΔM_x
ΔT_w
ΔP_x
ΔD_x
S_{ML}
(b) budget check, t = 550–650
level 14
level 15
0.00
0.02
0.04
0.06
0.08
0.10
0.12
mass residual
DNS momentum residual
ROM misfit left (E1/E0)